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Author(s): David Coleman
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The Twilight of the Census

DAVID COLEMAN

The lights are going out in census offices all over Europe. The traditional central source of population information, the demographers' rod and staff, the main source of data for planning for local authorities and central government, is being replaced by more timely alternative approaches, including population registers. In this regard, Europe is the world's demographic laboratory: the only continent where so many countries have adopted alternatives to the census (Valente 2011, p.14). It is only recently that all countries in the world have even held a census. At the other end of the scale, that source of data needed so badly by the least developed countries is being replaced in the most developed ones: yet another in the growing list of "demographic transitions" in which the data of demography itself are now the subject matter. It seems natural to regard this as an inevitable result of the coming of the electronic age, of the digitization, rapid transfer, and easy linkage of data, heralding a real revolution in knowledge and things knowable. But if so, why do some of the most advanced countries on Earth, including the richest and most powerful, retain their census, albeit with modifications? What may be lost, as well as gained, by this transition? And will the demographic convenience of modern methods of data-gathering come with a cost to freedom, as some fear?

Background to the census: From state exploitation and control to welfare and planning

These questions are not just of interest to demographers. The census has been entwined with the fundamental concerns of states ever since states arose. Earlier approaches to the counting of the people were seldom motivated by concern for their welfare, although the survival of the people and the survival of the state were ultimately synonymous. Compulsion was, and remains, central. If there were no enumeration without taxation, or if enumeration risked draft into the army or forced labor, the temptation was, and remains, to evade it. Correspondingly severe sanctions have been applied to defaulters, for example, in respect of those evading the old Chinese household register: "The family will be placed on the list of those liable for military service...any common people who hide from the census will be punished according to law and drafted into the army. If in their search the military come across minor officials who have suppressed the facts, those officials are to be decapitated" (cited in Cox 1970).

In time the census became less concerned with what the people could be obliged to do for the state and more concerned with what the state could do for them. By the late nineteenth century, questions on the condition of the people—housing, health, and disability—indicated the growing role of the census in social reform. Retrospective questions on children ever-born and children surviving, with a scale of social grading devised for the census as in the Census of 1911 in Great Britain, threw a spotlight on social and geographical inequality in infant mortality and (indirectly) on the onset of family limitation. As state spending increased in the developed world to between 30 percent and 60 percent of GDP, census questionnaires have grown in parallel. With the need for knowledge arising from local and national government responsibilities for schools and roads, educational qualifications, occupation, employment, and geographic mobility, the number of questions has inflated to 50 or more. And as populations have become more diverse through immigration, or as ancient minorities have asserted themselves, questions on identity have crept into the census, raising issues about the definition of populations (Kreager 1992).

The traditional census up to the 1970s

Up to the 1970s, all countries used the traditional form of enumeration: compulsory, comprehensive, total, periodical, instantaneous, household-based. Questions typically included details of the relationship of household members to one another, age, sex, marital status, education and employment, immigration and citizenship, previous residence. Fertility history, health status, journey to work, and car ownership may be asked; also racial or ethnic origin; rarely, income. Originally census questionnaires were delivered by an army of (mostly temporary) officials to each household and collected, checking that they had been completed. Now questionnaires are mostly posted out and posted back, or answered online. In the Czech Republic in 2011, 25 percent of forms returned were electronic, in Lithuania the majority, with back-up from enumerators. Post-enumeration surveys and other checks attempt to evaluate accuracy and missing data that could not be established by repeated calls. Missing data, even missing persons, are imputed by increasingly sophisticated statistical techniques (e.g., Abbott 2007). Until recently only the census could produce total population counts for the small areas that concern local authorities.

Reasons for abandoning the conventional census

Why do so many national authorities now look beyond the conventional census? Rapid population turnover, from accelerated internal and international migration, harshly highlights the inevitable delay in processing census results and their rapid obsolescence under modern conditions (*The Economist* 2010). International migration, always the least well recorded element of demographic change, is now its chief driver. Census data represent a statistical snapshot, taken once a decade, fixed in time in a world moving fast. That picture becomes distinctly sepia-toned as reality leaves it behind. And even on first appearance, the results show events as they were only in the recent past, not the present. With the most modern methods the first processed results seldom appear before a year's delay, and detailed analyses can take two years or more. Results of the (conventional) 2011 British census will not be complete for nearly three years (ONS 2012). The chief customers of the census, the local authorities, will have to wait for their data until March–October 2013. As British politician Harold Macmillan remarked as long ago as 1956 when Chancellor of the Exchequer, "some of our statistics are too late to be as useful as they ought to be. We are always, as it were, looking up a train in last year's Bradshaw" [timetable]. In promoting the American Community Survey, Leslie Kish, late chief of the US Census Bureau, remarked that the census "was great for its time, but not for today. Having censuses for local data once every 10 years is just not good enough anymore, and it hasn't been for some time." It may also be that coordinated repeat surveys organized by the EU such as SHARE, GGS, EVS, and others, despite their modest sample size and response rates, have persuaded politicians and bureaucrats that the census is less necessary.

Social and political developments have made censuses, and all inquiries by the state, less acceptable to some sections of the public. (And an incessant bombardment from unofficial "surveys" intended to extract money for charities and sales drives may have induced some fatigue.) Assurances that census information will be kept from the taxman and other official bodies may be disbelieved as trust declines while demand for greater accountability and democratic rights has increased (Warren 1999: chapters 4, 5, 6; Rose 1999). Some claim that a more geographically and socially mobile, less homogeneous society weakens trust and erodes social capital. In a supposedly "post materialist" world, the "second demographic transition" may threaten the data on which depends the evidence for its existence. In 2012 the annual Edelman Trust Barometer from 25 developed and semi-developed countries showed an unprecedented 9-point decline in the public's trust of government. Others claim that the basic premise is wrong, especially in Europe (Roger 2012). Libertarian political sentiments have gained ground in, for example, the US and Canada, objecting to almost any state data collection, and questioning its utility.

The state, as well as the people, has also begun to defect. Influential political voices have given added weight to popular mistrust, abolishing Canada's census long form and threatening the American Community Survey. As Joseph Chamie has observed (personal communication), "Most politicians prefer to remain ignorant. Without data, they can say crazy things and pass stupid policies and can't be challenged." In the UK Francis Maude, Cabinet Office minister in the 2010 Coalition government, has denounced the census as obsolete and inaccurate and hopes to save £480 million by cancelling the 2021 census—an unusually long-range policy for a politician (*Daily Telegraph*, 21 February 2011).

Writing about the 2010 debate on the Canadian census, Roderic Beaujot remarked, "It would appear that some of our political leaders consider that government can be based on values and good ideas, without the need for data and research. This is not unlike the view in some social sciences circles, that we should move to a post-empirical approach that is based on theory and qualitative approaches rather than empirical data" (Beaujot 2012, p. 1).

Even without considering ideological objections, social and economic change is making conventional enumeration more difficult in modern countries. Correct address lists are becoming difficult to construct. Some population groups are hard to contact or to count accurately. Young people are geographically highly mobile, especially students and young professionals. More people living by themselves complicate addresses for an occasional enumeration. An increasing number of people have two homes. Immigrants, especially illegal immigrants and overstayers, may be averse to government contact. Census questions duplicate some information already held in other official sources.

The census can be very expensive, especially if it relies on enumerators to deliver and collect forms and deal directly with respondents. Enumerators typically look after about 100 households each, so a large number of personnel must be recruited and trained. The US conducted a "shoe-leather" census until 1970. Even in 2000, 35 percent of households had to be visited in person by enumerators (the constitutional implications of the US census prohibit imputation of persons). Problems of cost and organization are exacerbated by the need to mobilize a large workforce every decade for a short time, rather than employing a smaller regular staff. And given the decadal interval, knowledge, experience, and expertise tend to be lost in the national statistical office as well. The average cost per head expected in the 2010 round of censuses in member states of the UN's Economic Commission for Europe was $8.84 (UNECE 2012, Table 1). The US easily tops the list at $48.90 per person for the 2010 census. Generally traditional enumerations are the most expensive. The register-based or mixed censuses of Austria, Belgium, Denmark, Finland, Netherlands, Norway, Slovenia, and Switzerland all cost less than $1.30 per head, Denmark's being cheapest of all at $0.03 (UNECE 2010, Table 4; expected costs on a purchasing power parity basis). However, costs are not easy to compare. The cost of a traditional enumeration census is one-off, not an annual cost for maintaining a set of registers. The cost per year falls to nearly zero in the middle of the intercensal period, rising to a peak in the census year in a kind of sine-wave pattern. The cost of a census based on registers will be high when it is set up for the first time, then diminishes over time (ONS 2012, Figure 2).

Replacing the army of enumerators by postal questionnaires saves money but may reduce coverage and accuracy. Many modern censuses only achieve contact with about 90 percent of their target population, and much less than that in areas with "hard-to-reach" populations. For example in the 2001 British census, in the area with the worst "census response rate," only 62 percent of the estimated population returned a form (ONS 2012). When the results of the census and the intercensal estimates do not tally, as often happens, there is no absolute yardstick to determine which is right, although the census used to be taken to be the gold standard. Intercensal estimates drift off-message as time passes, in recent years as a result of incorrectly estimated migration. In Germany, which until 2011 had taken no census since 1987, the population estimates were thought to be 1.3 million too high by 2010. On the other hand, the "missing million" judged to have been omitted from the UK census total of 1991 by comparison with intercensal estimates and other data (actually 1.2 million, Mitchell et al. 2002) was partly added to the total to bring it better into line with the population estimates. The results of the UK 2001 census overturned that correction, the blame now being placed on defective intercensal emigration estimates. Re-enumeration by the Census Coverage Survey and analysis and imputation of missing data and people were expected to give a compete estimate of 100 percent of the population to be covered—the so-called revision-proof "One Number Census." Inevitably, later local authority protests showed that claim to have been somewhat hubristic. The 2011 census total for England and Wales was 476,000 more than estimates based on the 2001 census; 200,000 of that was attributed to further error in the 2001 census total (ONS 2012).

Errors of up to 0.5 percent are regarded as acceptable, although discrepancies are more serious in the inner urban areas. Where numbers determine federal or other national central government subventions, lawsuits can drag on for years. The US government distributes almost $450 billion each year on criteria based on census results, and similar weighty sums follow census results in other countries. Naturally cities and local areas contest the results when they show a decline compared with the population estimates or the previous census.

What are the alternatives?

As state involvement in planning and welfare has expanded, so have the numbers of questions in censuses. The longer the questionnaire the higher the level of non-response and the greater the trouble and expense. To balance detail and coverage, some census bureaus have used a two-part inquiry whereby the majority of households received a basic shorter questionnaire, while a sample—usually 10 percent or 20 percent—received a long form with all the questions on which the government wanted answers. This was tried in the UK in 1961 but not repeated. These "long forms" can be very long indeed. In recent years about 17 countries have adopted this strategy, notably the US and Canada, but only Italy (2011), South Korea (2005 only), and Russia (2002 only) in the rest of the developed world. Elsewhere, it has been adopted by Bangladesh, Brazil, China, Ethiopia, Jamaica, Mexico, Nepal, Pakistan (1998), the Philippines (2000), Puerto Rico (2000) Sudan, and Vietnam (UN 2012).

In the US the long form introduced in 1960 was sent to one in six of the population asking details of socioeconomic status and housing conditions. In 2010 the short form really *was* short—just ten questions. The decennial enumeration has been updated since 2000 by the American Community Survey, covering a shifting sample in monthly rounds amounting to 3 million people per year (Mather et al. 2005). This rolling minicensus was described by former head of the Census Bureau Robert Groves as "one of the most valuable additions to society's knowledge about itself, because it's so current" (cited by Carl Bialik, *Wall Street Journal*, 30 March 2012). However, its compulsory status was revoked in the House of Representatives in May 2012 and the survey then declared unconstitutional, to the consternation of business interests and statisticians (Philips 2012; Rampell 2012). Its fate in the Senate is unclear in a dispute as yet unresolved (US Census Bureau 2002).

In Canada, where the census is held every five years, the short form was introduced in 1971, initially with 1/3 of households receiving the long form, 2/3 the short form. In recent years 20 percent of households received the long form. The short form comprised seven questions, the long form 47 plus eight on housing. By 2001 the long form had grown to 52, including questions on religion and birthplace of parents. By 2006, that had changed to eight questions on the short form and 53 on the long one. In 2011 the short form included the eight plus two more on language, still achieving a 98 percent response rate. But the long form had met its nemesis, abolished for the 2011 census by the Conservative government and replaced in 2010 by a voluntary National Household Survey (NHS) addressed to 4.5 million households. Denying that such a voluntary survey can substitute for mandatory census questions, Canada's Chief Statistician, Munir Sheikh, resigned in protest amidst a flurry of adverse media, statistical, and commercial comment (Dillon 2010; Beaujot 2012). The response rate to the NHS has fallen to 69 percent. Now even the short form is under Conservative attack for its "compulsory intrusiveness," and Statistics Canada is reported to be considering register-based alternatives (*Globe and Mail*, 5 May 2011). Politicians in North America seem to be marching in step away from comprehensive enumeration. In two major examples we see that political pressures on the census have driven the organizers to confine detailed questions to a fraction of all forms, only for the long forms to be challenged by further opposition to enumeration and abolished, or caponized by being made voluntary.

The rolling census

In France, they do things differently. There, a novel rolling census (*La Collecte Tournante*) has trundled along since 2005, replacing the conventional census last held in 1999. The rolling census is supplemented by an annual survey supported by an address register in the larger municipalities, the *Répertoire d'Immeubles Localisés* (Cézard and Lefebvre 2008; Desplanques 2008). A full census is conducted every year in one in five of the 35,750 small municipalities (fewer than 10,000 inhabitants). In the 900 larger municipalities (over 10,000 inhabitants) a sample survey of 8 percent of dwellings is conducted every year. Thus the total population in all small municipalities and 40 percent of the population of larger municipalities is covered over the five-year cycle—about 70 percent of the national population (Valente 2010) giving national information for each successive median year of the five-year cycle. As usual, the change was a response to demands for more timely data, a more even spread of costs over time, a lighter burden on respondents, and better quality control. This method does not reduce overall costs: each year's operations cost about one-seventh of a conventional enumeration (Pofantis 2008). But the advantages of quality and timeliness are great (Clanché 2011). No other country has adopted this approach although the UK ONS, obliged to find a census-substitute for 2021, is considering that among other options (ONS 2012, p. 10).

These developments in enumeration solve some problems but leave others untouched. Information is more timely but never covers all of the population at the same time. Individuals still have to answer questions. Problems of low response rates persist and no absolute yardstick of population size is created. This method yields no list of persons in the country who are obliged to contribute to the society or who are entitled to enjoy its privileges. Above all, it cannot cope any better than a conventional census with the powerful effects of international migration—the most dynamic of the three components of population change in most modern societies. Census-related methods can measure migration only indirectly. They make no use of the routine accumulation of administrative data, which have near-universal coverage, whereby the nation's employment, pensions, health, tax, and other encounters with the state are quietly recorded.

Population registers

Only population registers can do that. They offer a radical supplement, and possible replacement, for the traditional enumeration-based censuses. The utility of registers has been transformed by the huge explosion of information technology capability in the last thirty years, enabling population and administrative data to be entered, retrieved, compared, linked, and updated between corrected registers almost instantly, and population counts produced quickly. In some systems, total population counts can be provided more or less on demand. Of the larger countries of Western Europe, only France and the UK make no direct use of registers (other than address registers) in their enumerations.

The essence of a register is that it is current, being constantly kept up to date by additions and deletions through some form of record linkage. It is not merely a static list frozen at one point in time. A few countries have maintained a form of population register for centuries at the local parish or municipal level for the care of souls or for local civil administrative purposes, notably in the Lutheran Scandinavian countries (including Estonia). These were not initially in a form suitable for aggregation in a national population count and contained only the simplest information. As methods developed, as early as 1872 the International Statistics Congress recommended the general adoption of population registers (Brown 1872, pp. 447, 449).

By the beginning of the twentieth century population registers were operating in a number of modern and modernizing countries: Belgium, Chile, China, Czechoslovakia, Finland, Germany, Hungary, Italy, Japan, Korea, Liechtenstein, Luxemburg, the Netherlands, Spain, Sweden, and Switzerland. By the end of 1967 they were operating in at least 65 countries (UN 1969, table 1). Their modern (but not of course initially digitized) form took shape in the mid-twentieth century: in 1964 in Norway, 1967 in Sweden, and in some countries developed over time from simpler to more complex forms: from 1924 to 1968 in Denmark, from 1969 to 1971 in Finland, and from 1941 to 1972 in France. It was some time before these registers began to supplement, never mind replace, conventional censuses based on enumeration. Initially they were neither intended to serve nor capable of serving other than municipal-level functions, and the information they carried was elementary.

Almost all modern countries maintain other official registers of various kinds detailing the entitlements or obligations of their residents with respect to health and education services, taxes, license fees, and criminal records. These are not population registers; they are maintained for administration, not for enumeration. These "administrative sources" have come to embrace the majority, in some cases all, of the population as the supply and demand for services, welfare, and regulation involving the state has grown. Thus in the UK 95 percent of households are registered with the UK TV licensing authority; 72 percent of persons of eligible age are registered with the Drivers and Vehicles Licensing Authority. All in theory are now registered from birth with the National Health Service, and all over age 16 have a National Insurance Number. Perhaps the most interesting registers are those held by financial service companies. In the UK in 2010 there were 84.6 million debit cards in circulation, and 55.6

million credit cards in a total population of 61 million. In 2010, 90 percent of the adult population had a debit card, and 64 percent had a credit card. These facilitated 6.5 billion transactions in 2010 worth £410 billion (UK Cards Association 2011). Card companies hold substantial amounts of data about their subscribers, much more than many official registers.

Replacing the census

The Scandinavian countries of Lutheran heritage have obviously had a head start in these matters. From the eighteenth century, the Lutheran clergy were responsible for maintaining lists of households in their parishes as well as registering baptisms, marriages, and burials. Their societies are accustomed to registration systems and for the most part find their development uncontroversial. All four Nordic countries now have "censuses" that are wholly register based, starting with Denmark in 1981, followed by Finland in 1990 and Norway and Sweden in 2011 (Statistics Sweden 2006; Wullt 2008; UNECE 2007). The Netherlands held its last conventional census in 1971. Since then it has conducted a register-based "virtual census" supplemented by a survey (Prins 2000). Austria and Slovenia introduced register-based censuses in 2010 and 2011 respectively (Statistics Austria 2008, 2011), and 11 other European countries have adopted a mixed or hybrid system comprising register-based information supplemented by a "light" enumeration (Czech Republic, Estonia, Italy, Latvia, Lithuania, Switzerland), register data supplemented by a survey (Belgium, Iceland, Netherlands), or a combination of all three (Germany and Poland) (United Nations 2010, 2012). The former West Germany has had no census since 1987, the former East Germany not since 1981. The German 2011 combined census is evaluating the data from administrative registers (local population registers and the Federal Employment Agency register), from detailed questionnaires sent to 17.5 million owners of buildings and dwellings, and the results of a sample survey of about 10 percent of all residents (Augustyniak 2011). This unique arrangement was made necessary by the absence of any centralized registers of population or of dwellings in Germany, reflecting sensitivities to past politics. Registration in the communal (local) population register is compulsory, however. That includes, among others, the names, sex, date and place of birth, marital status, citizenship, place of residence, and address of each person. The 13,500 local authorities use standardized systems. Most respondents were not involved directly in this operation, an important consideration in Germany and in register-based operations generally.

In the 2010 round, 21 out of 40 countries in the UNECE still used the traditional form of census, mostly countries in Southern and Eastern Europe (most of the area east of Hajnal's line; see UNECE 2010, Tables 2a, 2b; Valente 2010, map 1), plus Canada and the United States. Five countries adopted a register-based approach entirely; the remaining 11 adopted a variety of approaches. Between the 2000 and 2010 rounds, nine countries changed their form of enumeration: six to a combined method (Czech Republic, Estonia, Germany, Italy, Lithuania, Poland), two to a register-based census (Austria and Sweden 2011), and one to a rolling census (France).

Drawbacks of register-based enumeration

But there are major imperfections in the data-gathering capacity of register-based censuses, and their development also raises broader political, social, and technical issues. Register data are peculiar in that data supply and demand operate the wrong way round. Normally, in censuses and surveys, the organizers determine what information is needed and devise questions to elicit those data. With a register-based census, the nature of the administrative sources dictates what can be provided, and it may well have been devised for purposes that differ from the usual requirements of census takers. They are set up for administrative, not statistical purposes. The quality of registers may be uneven and may initially at least be the responsibility of different organs of local or central government. Data deemed of lesser importance for administrative purposes will receive less attention—for example on heating and water supply in the Norwegian housing register. Data on events that happened before the establishment of the register or on events occurring abroad will be absent. In most countries cohabitation is not registered except incidentally in relation to tax or custody of children.

Data that should be identical in different registers may not be, and, for linkage, individuals need to have the same person-number in the same format. Making a set of registers fit for the purpose of substituting for census can take many years. For example, the Central Population Register in Finland was set up in 1969 and the eight other registers (business, dwellings, housing, education, employment, family, household, and income) came on stream at various later times, the latest being employment in 1987. While they began to contribute to the census a few years after their creation, they were not all fit to contribute together to the first wholly register-based census until 1990. All this involves high initial set-up costs although, once up and running, costs will be low and real savings can be made.

Some phenomena of interest are typically omitted by registers. For example, German administrative data do not contain information on self-employment, education, or migration background (that is, birthplace or citizenship of the respondent's parents). There are no complete registers of buildings and dwellings, or of people living in residential homes or other communal institutions. Separate surveys must supply that information.

The Austrian registers lack data on occupations. That is unusual. But like most registers they also lack information about the journey to work or to school or the means of transport. "Activities" as opposed to "attributes" tend not to be recorded. Information on persons available for work and on jobseekers is also generally lacking, and on unemployed persons who are not registered as unemployed (e.g., Statistics Norway 2008; Statistics Netherlands 2012). "Households," defined as persons living in the same dwelling with joint board, cannot easily be reconstructed using register information (Statistics Norway 2008, p. 6).

The questions in a conventional census are altered from time to time to meet emerging information needs in the society, for example on use of information technology, and to illuminate the new salience of disability and diversity. The census can respond flexibly to new needs simply by inserting new questions. It is more difficult to change registers in that way. In practice, surveys are used in some countries to supplement data missing from register-based censuses, although only on a sample basis. In the Netherlands "virtual census" of 2001 and 2011, register data were integrated with information from the regular Labour Force Survey to provide data, absent or incomplete in the registers themselves, on economic characteristics (occupation, unemployment, educational attainment).

Registers do not wholly solve the problems of keeping tabs on migration unless arrivals and departures are linked to the population register at the border. Out-migrants or emigrants may forget to cancel their registration. When people leave permanently without canceling their registration, in theory they would "live" forever. Illegal immigrants and overstayers may evade the registers. But in the countries discussed here it is almost impossible to live a normal life without registration, since public services are thereby inaccessible.

For a variety of reasons, then, a number of European countries that have abandoned the traditional census have not yet gone the whole distance to an entirely register-based survey (Statistics Netherlands 2012). In all, 13 countries adopted a mixed approach in the 2010/11 census round, combining register data either with a "light" enumeration or with survey data, whether routine (from the Labour Force Survey) or specially constructed for the event. Six of them made this switch for the first time (Czech Republic, Estonia, Germany, Italy, Lithuania, and Poland), as did two countries outside Europe (Israel and Turkey). Six others had used this method in 2000 as well (Belgium, Latvia, Netherlands, Slovenia, Spain, Switzerland; UN 2012).

Broader implications

So much for the detail. What are the broader implications of the movement away from comprehensive decennial censuses, with their 40 or 50 compulsory questions, toward a wholly or partly register-based system where an individual's profile is on permanent and in some aspects unalterable official record? Is anything lost in this exchange? The statistician William Kruskal (1984) felt that the US decennial census, with all its publicity and its (theoretical) involvement of all residents, was a major national "ceremony" of importance, the only time when all residents were involved simultaneously with a major official activity, asserting their presence as US residents for the record. In the US, this is given added significance by the Constitutional status of the census. In Canada, too, statisticians have risen up to defend the census from parliamentary attack, noting that "participation in statistical enquiries is an essential part of citizenship in a modern democracy" (Dillon 2010) and, echoing Kruskal, that the census has a "symbolic value, not unlike history, literature, maps and museums in defining ourselves as a country, ... evidenced in the media interest in census results" (Sheamur 2011, all cited in Beaujot 2012). In Europe, where opposition to census taking and other official inquiries has been growing, such affectionate attachment to the traditional census is difficult to imagine although it certainly has its defenders (Boyle and Dorling 2004).

Although many find census questions irksome, few now fear that a response will involve them in cost or obligation. In developed countries census results are ring-fenced from other agencies of government, especially taxation authorities. Individual questionnaires are usually kept secret for 100 years, and small-area breakdowns of published data are randomly anonymized to prevent individuals being identified. Generally speaking, the public in most countries has had faith in the privacy of the census; otherwise non-response would be much higher than it is.

Registration has different implications. Here the record is permanent. It follows individuals through time and space, logging their major life events. It is difficult to evade. In its most developed form the person-number of the population register links to tax and medical records. Big Brother may be watching you, some fear, and rather efficiently thanks to modern IT. Administrative sources provide plenty of opportunities for silent data-gathering, with little involvement or awareness by citizens, potentially more complete and comprehensive than any census. Exceptional measures are needed, in democratic countries, to ensure that information does not leak from those official agencies permitted by law to access the data (Watner and McElroy 2003). For example, the local registers of the Netherlands are connected with cut-outs, with an electronic one-way valve such that data may be sent from them to Statistics Netherlands, but the latter cannot interrogate the local registers (Prins 2000). Elaborate ring-fencing arrangements between registers provided by government agencies also exist in the Austrian and other systems. No serious problems have been reported in European countries using registers, although these are all established democracies with well-developed self-consciousness concerning human rights. But it was inauspicious that in 2011 six people were arrested for stealing and distributing the entire Israeli population registry data on 9 million persons.

Politics and culture matter greatly in determining public attitudes toward censuses and registers. Countries where registration has been a fact of social life for centuries, as in Scandinavia, and where this system has so far been demonstrably harmless, appear to have had few problems in coping with the transformation of their traditional registers into machine-readable form and eventually integrating them to create a census substitute. But past oppression has cast a long shadow. The extension of German census questions to religious and racial origin had disastrous results for Jews and other population groups beginning in 1933 (Aly et al. 2004). But the listing of Jews and other victims was performed initially through the census data of 1933 and 1939, not through any existing register. These served to create lists maintained by the Nazis. It is the political culture that determines the problem, not the method of gathering data nor even the nature of the data collected. Furthermore, it does not require a census or register to identify people who are distinguished by appearance, surname, or area of residence. The events up to 1945 have had a lasting effect on German attitudes toward data collection, and on those of Jews everywhere except in Israel. Paradoxically, Israel has had a population register since 1948, computerized in 1965, using a person-number system in which religion, ethnic group, and nationality are recorded (National Center for Health Statistics 1980; Zadka 2008). Israel also controls, and has "frozen," the Palestinian population register, which determines who may lawfully reside in the West Bank (Human Rights Watch 2012).

Lights in census offices continue to burn bright east of Hajnal's line. There only the most Western and Westernized countries (Estonia, Slovenia) have adopted registers for the census. The English-speaking countries, Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States, all still depend on traditional censuses, with some experimentation in North America. Until recently none but the UK had considered using administrative data for census purposes, but political opposition to traditional information-gathering is pointing the way to registers in Canada too. In the UK an ambitious program to create an identity card backed by a limited register, proposed by the last Labour government to improve security and the control of immigration, met strong resistance (Watner and McElroy 2003; Enterprise Privacy Group 2005). It was scrapped by the incoming Conservative–Liberal Democrat coalition government ostensibly on grounds of costs and intrusiveness, although some denounced its removal as mere political opportunism, exploiting a difficulty for the governing party. There had been precedents: identity cards and a simple national register had been created in wartime in 1915 and 1939, the latter surviving seven years into peacetime only to be abandoned in 1952, again by an incoming Conservative government. The latter register formed the basis for food rationing and, later, the NHS number.

In the Anglosphere, elements of the libertarian and traditional Right and the liberal Left find common cause in suspicion of the census, regarding suggestions of a population register as a further intrusion by big government. Census questions have been probing deeper into identity—first ethnic origin, then religion. In the UK questions on income and on sexual orientation were considered for 2011 but rejected. In reaction to census "mission creep," some journalists on the Right have suggested that the UK census be "pared back to its original purpose: a simple population count" (Johnson 2011). Francis Maude, Minister responsible for the census, announced a decision to scrap the census in favor of new ways of counting the population that would be "less intrusive." There has been no tradition of keeping population registers in the English-speaking world, which shares a Common Law tradition, not the codification of statute typical of Continental Europe. Official identity documents are regarded as an undesirable infringement on freedom, something best confined to the less happy lands of Continental Europe. Some English people take perverse pleasure in having no such documents, even though ordinary life in Britain, especially financial transactions, cannot get far without a bank card, a driving license, a national insurance number, or a diary full of passwords. Similar attitudes seem to have been inherited by the countries of the Anglosphere with their traditions of even more robust independence and individualism.

If the case for population registers were to make progress in the English-speaking countries, major questions would arise on the future of information on ethnic and religious identification and identity. The UK and all the English-speaking countries abroad have developed comprehensive sets of self-ascribed questions on ethnic identity, ancestry, religion, and language. These questions have proliferated with successive censuses, in line with policy responses to the new diversity arising from large-scale immigration. In the UK they have become pervasive for the enforcement of equality legislation and its quotas and targets—routinely posed for application to school and university, for recruitment, promotion and contracts in the public sector, and for services provided by the National Health Service and by local authorities down to planning permission for kitchen extensions. In Eastern Europe, where nation-states have only recently emerged from ancient empires, these questions apply mostly to long-standing indigenous minorities. In between, in Western Europe such questions (except, in some countries, religion) are not asked; multicultural policy has not been developed. Instead, registers allow the enumeration of "foreign background" populations derived from the birthplace / citizenship of individuals and their parents, as in Denmark, Netherlands, Norway, Sweden, and some others. In France, however, the official collection of such data is held to be incompatible with the equality of citizenship, and surveys suggest that public opinion is very uneasy about using "ethnoracial" classifications in any inquiry (Simon and Clément 2006). Most Southern European countries collect information only on birthplace and on nationality.

Any move in the English-speaking countries toward registers for enumeration would fall foul of these policies. Register data suit fixed attributes of individuals (date of birth, sex, educational attainment) or characteristics that alter relatively slowly and in an objectively verifiable manner (occupation, address), both of which are routinely recorded administratively. They are unsuited to self-ascribed characteristics divided into many different categories, or specified exclusively and at will by the respondent independently of any official list. Would the 420,000 who returned "Jedi" as their religion in the UK census in 2001 choose to enter that into a register? Ethnoracial or religious identity on a register would endow that person with a (potentially) permanent ethnic or racial label. The consequent racialization of society, making ethnic divisions in society permanent through statistics and reinforcing self-conscious separate identity, would be unlikely to promote common citizenship and social cohesion.

It is difficult to envisage such a development in a democratic society. The prospect might provoke a reassessment of ethnic statistics. Basic aggregate information could continue to be gleaned from survey questions. Registers without personal ethnic or religious labels would be a first step away from what many regard as a disturbing trend of social fragmentation, where statistical categories structure society rather than reflect it. Similar concerns have been raised elsewhere, for example regarding the proliferation of ethnic and color questions in the census of Canada (Beaujot 2012).

Where do we go from here? In Continental Europe at least, the trend toward the use of registers to supplement or replace the census is clear and probably unstoppable. Perhaps few countries will rely on registers alone. The favored approach in 2012 is to supplement register data with a survey to provide data that registers cannot reach. Whether that is just a half-way house to wholly register-based censuses remains to be seen. We may have passed the high-water mark of information-gathering about populations, and the price of further progress down the road to the register census will be fewer demands for personal information, just as the means to process it and link it grow.

As the world develops and IT capacity is globalized, countries outside Europe in the developing world are creating electronic population registers (e.g., Brazil, Malaysia, Singapore, Thailand, Turkey). Europe and the West may well be overtaken in the modernization of population data as they have been in economic growth. Not all these projects have respectable antecedents. The modern Chinese household registration system (*hukou*) (Chan and Buckingham 2008) is the lineal descendant of the world's oldest. That existed as an inventory of population for state purposes and to control internal migration. It still does the latter, attempting to separate the rural and urban social and economic spaces, cutting off rural migrants to cities from urban entitlements. The United Nations noted (2005, p. 3) "with deep concern the *de facto* discrimination against internal migrants in the fields of employment, social security, health service, housing, and education that indirectly result[s], inter alia, from the restrictive national household registration system (*hukou*) which continues to be in place despite official announcements regarding reforms."

The world's biggest registration project, in democratic India, serves different ends. This astonishingly ambitious project aims to register all residents of India by 2021, including illiterates, and provide them with an identity card with biometric data (Office of the Registrar-General India 2009). One of its chief advantages echoes the ancient benefits of the parish registers: an official record of name, existence, and parentage, therefore of entitlement (to benefits, to property, to vote) and of inheritance. Millions of impoverished Indians, lacking documents, are denied the ability to resist exploitation and exercise claims on the state. That is an instructive contrast of the divergent purposes to which registers and enumeration may be put in different kinds of political regime: some benevolent, as in this case; others malign, as in Germany in the 1930s.

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